CHAPTER QUESTIONS

SKILLS INTERPRETATION 7

1 a Draw the structural formulae for the following compounds:

i ethyl ethanoate

8 ii butyl butanoate

7 b Draw the displayed formulae for the following compounds:

i methyl butanoate ii ethyl propanoate

SKILLS CRITICAL THINKING

c Name the following compounds:

i $CH_3CH_2COOCH_3$ ii $HCOOCH_2CH_3$

SKILLS INTERPRETATION 8

2 Suppose you have access to bottles of ethanol, propan-1-ol, ethanoic acid, propanoic acid and concentrated sulfuric acid.

a Draw the structural formulae of all the esters you could make from combinations of the alcohols and carboxylic acids.

SKILLS EXECUTIVE FUNCTION 5

b Describe how you would make small samples of each of these esters.

6 c Explain how you could detect the formation of the esters in these reactions.

SKILLS CRITICAL THINKING 5

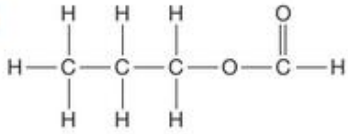
d Give two common uses for esters.

SKILLS CRITICAL THINKING, INTERPRETATION 8

3 Give the names and draw the structural formulae of the alcohols and carboxylic acids that could be used to make the following esters:

a methyl propanoate

b



c butyl ethanoate

SKILLS INTERPRETATION 7

4 Ethyl ethanoate can be made from ethanol as the only organic starting material in a number of steps.

a Draw the displayed formula for ethyl ethanoate.

SKILLS CRITICAL THINKING 5

b Identify the carboxylic acid that is required for the esterification of ethanol to form ethyl ethanoate.

c Describe how you could make the carboxylic acid in b from ethanol.

SKILLS PROBLEM SOLVING 9

d Write a balanced chemical equation for the formation of ethyl ethanoate from ethanol and the carboxylic acid in b.

SKILLS CRITICAL THINKING 5

e Write down the conditions required for the esterification step.

SKILLS EXECUTIVE FUNCTION 6

5 If you were given three unlabelled samples of ethanol, ethanoic acid and ethyl ethanoate, but weren't allowed to smell them, suggest how you might identify which one was which, with as much certainty as possible.

END OF CHEMISTRY ONLY